

CH365 Chemical Engineering Thermodynamics

Lesson 19 Heat Effects of Industrial Reactions

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CANVAS Modules, Lesson 19

Industrial Reactions

- Rarely carried out under standard conditions
- Reactants not in stoichiometric proportions
- Reaction may not go to completion
- Final T may differ from initial T
- Presence of inerts
- Multiple reactions

Review of Lesson 18

Today we will use these in the homework.

$$\Delta H^\circ = \Delta H_{298}^\circ + R \cdot \int_{T_0}^T \frac{\Delta C_p^\circ}{R} dT = \Delta H_{298}^\circ + R \cdot \text{IDCPH} \quad \text{Eq. 4.19}$$

$$\text{IDCPH} = \int_{T_0}^T \frac{\Delta C_p^\circ}{R} dT = (\Delta A) \cdot (T - T_0) + \frac{\Delta B}{2} \cdot (T^2 - T_0^2) + \frac{\Delta C}{3} \cdot (T^3 - T_0^3) + \Delta D \cdot \left(\frac{T - T_0}{T \cdot T_0} \right) \quad \text{Eq. 4.20}$$

This only works when T is the same for reactants and products. (see L18 Slide 5)
Not explained in book!

$$\Delta A = \sum_i \nu_i \cdot A_i, \quad \Delta B = \sum_i \nu_i \cdot B_i, \quad \text{etc.}$$

$$\Delta H^\circ = \Delta H_{298}^\circ + R \cdot \int_{T_0}^T \frac{\Delta C_p^\circ}{R} dT = \Delta H_{298}^\circ + R \cdot \text{MDCPH} \cdot (T - T_0) \quad \text{Eq. 4.22}$$

$$\text{MDCPH} = \frac{\langle \Delta C_p^\circ \rangle_H}{R} = \Delta A + \frac{\Delta B}{2} \cdot (T + T_0) + \frac{\Delta C}{3} \cdot (T^2 + T_0^2 + T \cdot T_0) + \frac{\Delta D}{T \cdot T_0} \quad \text{Eq. 4.21}$$

Functional nomenclature used in book, page 155

$$\int_{T_0}^T \frac{\Delta C_p^\circ}{R} dT = \text{IDCPH}(T_0, T, \Delta A, \Delta B, \Delta C, \Delta D)$$

(no equation numbers)

$$\frac{\langle \Delta C_p^\circ \rangle_H}{R} = \text{MDCPH}(T_0, T, \Delta A, \Delta B, \Delta C, \Delta D)$$

These are “parameters;” all but T_0 change

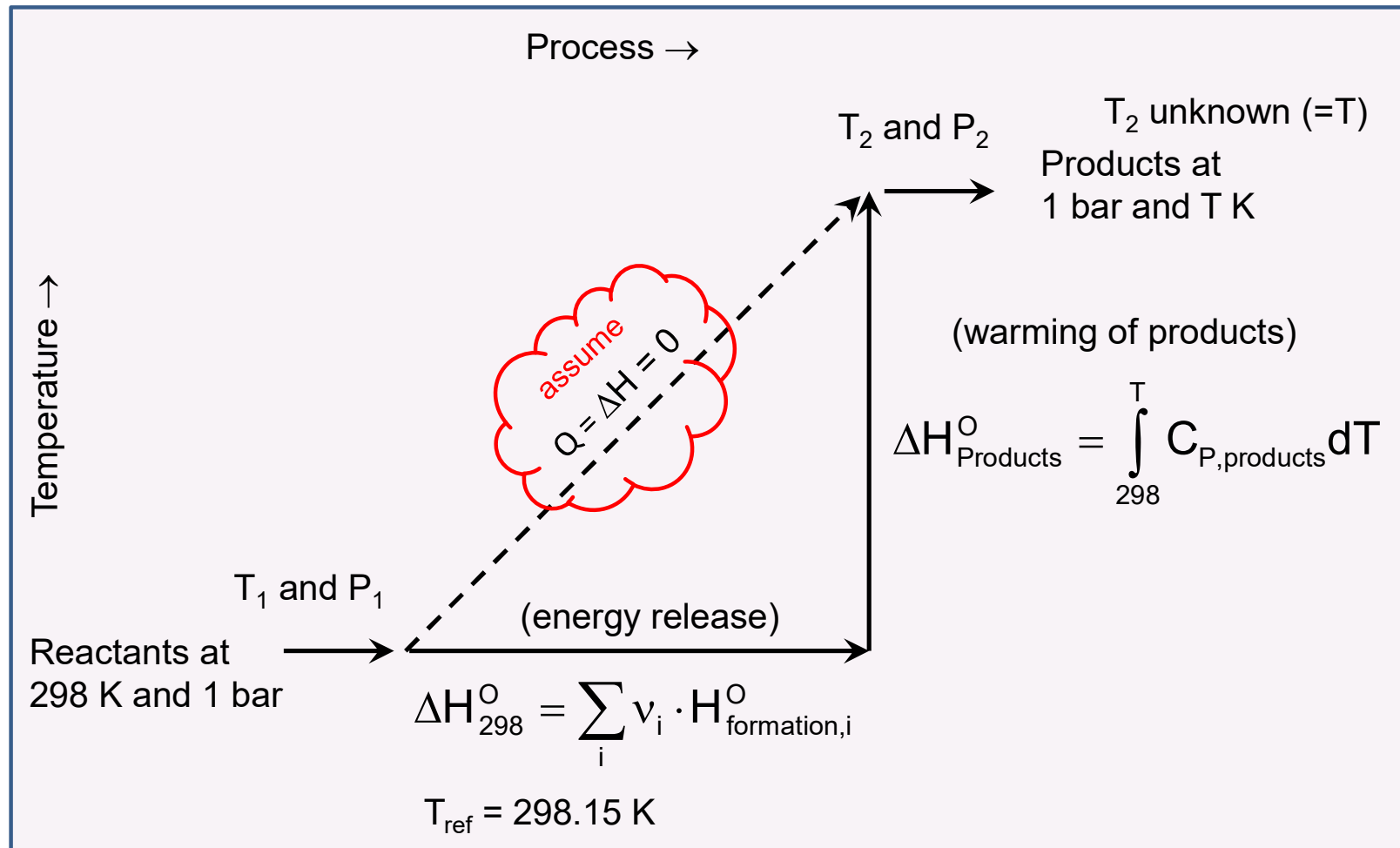
Example 4.7

What is the maximum temperature that can be reached by the combustion of methane with 20% excess air? Methane and air enter the burner at 25°C.

Assumption: Rate of combustion is much greater than the rate of heat transfer.

This means that the heat buildup in flame is essentially adiabatic.

Important Example. “Adiabatic Flame Temperature”



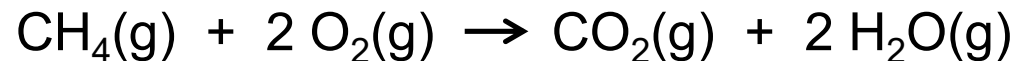
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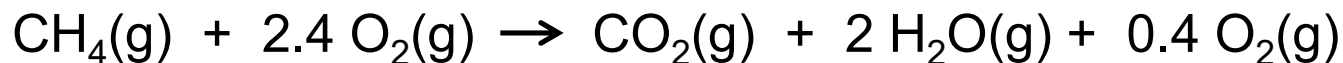
Solution:

Part (1) Calculate heat of reaction at 298:

Balance:



Add 20% excess O_2 :



Add N_2 :

$$\text{N}_2 \text{ in feed} = (\text{O}_2 \text{ in feed}) \cdot \frac{79}{21} = 2.4 \cdot \frac{79}{21} = 9.03 \text{ moles}$$



Questions?